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Networking practices in design research: refining design principles

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In a design research (DR) study concerning how students exercise their representation competency when using digital tools, we aim to investigate the potential of networking practices to support the purpose of developing theoretically grounded design principles in DR. We provide and discuss an empirical example of how design principles for students' activation of representation competency, when using GeoGebra, are adjusted through retrospective analysis in DR. We further show how such analyses may be structured through the coordinating strategy of networking of theories.

Keywords: Design research, networking of theories, predictive theory elements, design principles.

Introduction

DR aims to develop both theory and practice in an intertwined manner. An important part of DR is to develop principles for designing tasks through careful analyses using theoretical perspectives (Prediger, 2019). However, the development of theory only appears to a limited extent in some DR studies (diSessa & Cobb, 2004). Networking of theories offers strategies for understanding and relating theoretical constructs, while at the same time aiming to connect the many unrelated theoretical perspectives in mathematics education (Prediger et al., 2008). In this paper, we aim to investigate how practices from networking of theories may strengthen theory development in DR. To achieve this, we use an illustrative case involving task design aiming to exercise students' mathematical representation competency when using digital tools, such as dynamic geometry environments (DGE)). The Danish competency framework (KOM) defines eight competencies, among them is the *representation competency*, which involves the capability to make use of, reflectively choose, interpret, translate between mathematical representations and understanding the scope, limitations, and strengths of representations being used (Niss & Højgaard, 2019). For instance, students exercise representation competency when choosing the most appropriate representation for given phenomena or understanding relations between representations (e.g., graphs and equations). KOM does not address the use of digital tools at large, why coordination of two fine-grained theoretical perspectives is relevant. For the use of digital tools, we apply the instrumental approach (e.g., Trouche, 2005), and for further insight on the mathematical representations, we utilise Duval's (2017) perspectives on semiotic registers. Our research question is *Applying practices from networking of theories, how may design principles about students' exercise of representation competency when using digital tools be refined*? We present our DR process focusing on humble design elements. Then, we present perspectives on networking of theories and our theoretical perspectives, followed by our illustrative case of which we carry out parallel analyses conducted to define our design principles for exercising representation competency while using digital tools based on the humble design elements. Finally, we refine the design principles using a networked analysis in a discussion and networking practices in DR.

Design research and importance of design principles for task design

DR includes different phases: 1) *preparation of design*, involving the development of design principles; 2) *testing design* by gathering data in classrooms; and 3) *retrospective analyses*, which serve to analyse the design and refine the design principles (diSessa & Cobb, 2004). Theorising exits in all steps of DR. Different theory elements in DR may be characterised as categorical, normative, explanatory, descriptive and predictive theory elements. In particular, *predictive theory elements* are key for this paper, as it involves the development of *design principles* (Prediger, 2019). Design principles may have the following structure:

> If you want to design intervention X [for the purpose/function Y in context Z], then you are best advised to give that intervention the characteristics A, B, and C […], and to do that via procedures K, L, and M […], because of arguments P, Q, and R (van den Akker, 1999, p. 9)

Based on a previous literature review (Pedersen et al., 2021), involving the representation competency of KOM (Niss & Højgaard, 2019) and the perspective of semiotic registers (Duval, 2017), we present the *humble predictive heuristics*, meaning the design principles that have not yet been tested. The review revealed that students could engage in processes of dealing with different representations and their reciprocal relations in a multi-representational digital tool. Yet, there is a risk that the digital tools outsource essential aspects of mathematical activities by automatically generating translations between representations. As part of the literature review, we proposed five recommendations for task design aiming to have students exercise their representation competency in interplay with digital tools (Pedersen et al., 2021). Three out of the five recommendations are relevant for this study. According to van den Akker's (1999) definition, we regard them as *humble design characteristics* (HDC):

- HDC 1. Including the linguistic register by formulating the task in natural language and asking for answers in natural language.
- HDC 2. Breaking objects and windows of the tool into smaller units by introducing them gradually.
- HDC 3. Using sliders, dragging and tracing for explorations of different representation and representation forms' reciprocal relations.

Networking of theories and presentation of selected theoretical perspectives

Networking of theories provides strategies for connecting theories at different levels: From 'understanding others'/'making understandable' to 'integrating locally'/'synthesising'. We focus on the strategies 'combining' and 'coordinating', both focus on understanding an empirical phenomenon. Often, more than one theory is necessary to understand an empirical problem. When 'coordinating', the involved theories must be *compatible*, meaning that their *background theories outside mathematics education* (Bach et al., 2021) are not in conflict (Prediger et al., 2008).

We aim to network theoretical perspectives to understand the interplay of students' representation competency and their use of digital tools, which are compatible with KOM. KOM plays the role of what Bach et al. (2021) term as a 'framing theory', meaning a background theory inside mathematics education research. Although, it may rather be a framing framework. KOM is of a cognitive nature focussing on an individual's expression of cognition in particular mathematical situations (Niss &

Højgaard, 2019). Semiotic registers and the Instrumental Approach are in line with KOM as both address cognition. As the theoretical perspectives look at different objects and phenomena (mathematical representations and use of digital tools, respectively), they may complement each other. Hence, their backgrounds appear not to conflict indicating that 'coordination' may be possible.

Transformations between semiotic representations are essential in mathematical activity. In their nature, mathematical objects and processes are abstract and only accessible and transformable through their representations. Consequently, the process of mathematical activity involves transforming representations within or between registers. Four registers are distinguished depending on if a register is multi- or monofunctional and discursive or non-discursive (Duval, 2017). Discursive registers (the *symbolic* and the *linguistic* registers, named as such in Pedersen et al., 2021) involve written and spoken languages expressing meaning units of thoughts. Non-discursive registers (the *graphic* and the *figurative* register named as such in Pedersen et al., 2021), only display one (or more) static, visual object(s). Mono-functional registers (the *sy*mbolic and the *graphic* registers) involve algorithmic processes. Multifunctional registers (the *linguistic* and the *figurative* registers) then involve non-algorithmic processes, such as communication and imagination. Each register may contain more representation systems with their own rules. A transformation between representations within the same register is called *treatment.* A transformation between representations from one register to another is called *conversion* (Duval, 2017).

The instrumental approach to mathematics education concerns the cognitive process of turning an artefact (e.g., a ruler, calculator, or dragging in GeoGebra) into an instrument for the activity of solving any kind of task. When distinguishing between artefact and instrument, it is inherent that the artefact in itself is just a material object, and only becomes an instrument when an individual can appropriate parts of the artefact for a certain task. This process is called *instrumental genesis* and is developed over time and shaped through the two-way processes of instrumentation and instrumentalisation. *Instrumentation* is directed from the artefact towards a user, as its constraints and possibilities shape how an individual uses the artefact. Instrumentation can go from a phase of exploiting different possibilities of using the artefact to solve the task relying on prior knowledge to a more stable phase, where the same usage is applied for similar tasks (Trouche, 2005). *Instrumentalisation* is directed from the user towards the artefact, as the user uses and manipulates the artefact (based on the user's knowledge), which can be different from the original intent of the designer of the artefact and can both lead to enrichment of an artefact, or its impoverishment. Within this two-way process of instrumental genesis, an instrument involves an artefact (a material thing) and schemes concerning the mathematical content and its relation to the use of the artefact (Trouche, 2005).

Presentation of the case: Ivy and Aya

The data excerpt and task presented in this paper were chosen because the task represents humble design characteristics. The task (inspired by Johnson & McClintock, 2018) was originally developed to activate the communication competency, which often also involves one or more mathematical representations (Niss & Højgaard, 2019). The task relates different representations and different mathematical contents (geometry and functions) in a GeoGebra template. Further, in this paper, we

relate the task to the representation competency. In the graphic view (i.e., the graphic environment in GeoGebra), a rectangle is constructed to the left of a coordinate system (first quadrant) with a point *P*. Point *A* of the rectangle can only be dragged vertically, such that the height changes and point *B* and C are fixed. Point P's *x*-coordinate corresponds to the rectangle's height and the *y*-coordinate to the area of the rectangle. When dragging point *A*, point *P* moves on the line $y=3x$. Unintentionally in the task design, it is also possible to drag point D on the rectangle horizontally. When dragging point *D*, point *P* moves on the vertical line, $x=|AB|$. The algebra view is also accessible in the template. The task and the transcript are translated from Danish. The task has four phases: 1) Investigate the rectangle and features related to it without dragging; 2) investigate the relationship between the rectangle and point *P* by dragging point *A*; 3) fill a table for related values; and 4) represent the functional relationship (to be $y=3x$), by its formula and a graph. The following excerpt presents two students' dialogue about the relationship between point *P* and the rectangle in phase 2, see Figure 1.

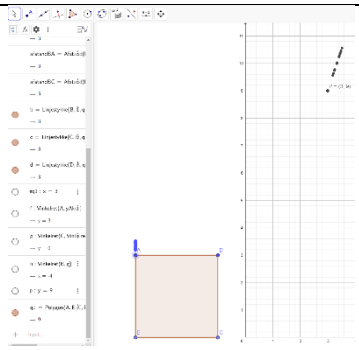
Investigate the construction on your computers by dragging point A in the figure. Describe the relationship between point P and the shown figure. For point P, describe what characterises the x-coordinate and the y-coordinate. Explain why the relationship exists.	

Figure 1: Task and illustration of the template with traces (Johnson & McClintock, 2018)

Before discussing the task, the students investigated the template individually. Prior to this excerpt, the students explored the dragging possibilities, e.g., dragging points *A* and *D*. Dragging point *D* was neither intended nor part of the task description. The students began to discuss, dragging point *A*:

1	Ivy	The area moves *P*. Okay, the area moves *P* up.
2	Aya	No, it is not the area. You mean the width.
3	Ivy	Yes, but the width changes the area
4	Aya	So does the height…
5	Ivy	Yes, yes.
6	Aya	But both factors move the area
7	Ivy	No, but, no… Try to see… try to see. If you move this one here, it changes both the area and the height, then it [point *P*.] moves askew. [Ivy drags point *D* back and forth, slowly. The figure is no longer a rectangle.].
8	Aya	Yes, yes, but Ivy… I know that both elements depend on the area. The area is the height times the width.
		[...]
9	Aya	What have you written?
10	Ivy	I have written that if you change the rectangle's height to 0, the intersection *P* ends at (0,0). I think that the intersection is determined by the rectangle's height and area. The area moves *P* up, up by the *y*-axis, and the height moves it by the *x*-axis. When you change the height, you also change the area and *P* therefore moves askew away from (0,0).

Line 10 shows that Ivy concludes on the relationship when dragging point *A* (not *D*), as point *P* moves "askew" and not vertically.

Analyses of the case

To 'coordinate', we analyse data in parallel, meaning that we analyse using (1) semiotic registers (Duval, 2017), and (2) the instrumental approach (Trouche, 2005). Then, we 'coordinate' the analyses (Prediger et al., 2008). We have regarded the students' meanings and productions as collective, hence, considering them as a pair rather than individuals.

Analysis through semiotic representations: Ivy and Aya orally discuss the relationship between point *P* and the rectangle. First, they characterise what defines the area, that is the width (line 3) and the height (line 4). Secondly, Ivy drags point *D*, performing treatments on the quadrilateral, while the students do conversions between height and area of the rectangle to point *P* (line 7). Thirdly, they connect the height with *P*'s *x*-coordinate and the area with *P*'s *y*-coordinate, which is specified at the end (line 10). Finally, the students use the word "intersection" in their natural languages in line 10 in connection with point *P*. It is unclear how the students interpret/understand "intersection", but it appears to stem from the description of point *P* in the algebra view. Possibly, they use it as a description of point *P* as they say "the intersection *P*", meaning where the height of *AB* and the area intersect.

Analysis through instrumental genesis: Prior to the excerpt and at the beginning of the transcript (lines 1–6), both Ivy and Aya explore the GeoGebra template using their previous knowledge of rectangles. As they drag point *D*, point *P* does not follow the intended line (line 7). The possibility of dragging point *D* is a constraint of the task related to the process of instrumental genesis, particularly for the instrumentation process, when the students explore the tool. This is reflected in their actions (see line 7). As part of their instrumentalisation, they use GeoGebra for what they believe was the intention of the task and realise (see line 10) that they should drag point *A*, which is the intent of the task. The term "intersection" appears in the algebra view, and may consequently be regarded as a constraint of the tool: it leads the students to use a term that stems from the tool but is inappropriate in the given situation. This may even lead to an incorrect understanding of point *P*. To use dragging as an instrument for this particular situation, Ivy and Aya should be able to find the relation between the rectangle and point P while dragging. Hence, the instrument of dragging for Ivy and Aya is developed for answering the task and the instrument relates both dragging (i.e., the artefact) and schemes relate the content of linear functions and geometrical knowledge on rectangles with the specific actions of dragging.

Summary of analyses: 'Coordinated' analysis of Ivy and Aya working with the task

When Ivy and Aya explored the template using prior knowledge on rectangles (lines 1-6), it involved understanding and making conversions between and treatments within the involved registers (i.e., the linguistic, the graphical and the figurative) by relating and expressing what defines the area. Dragging point *D* created constant treatments of 'rectangle' *DABC* to be a quadrilateral. For the instrumentation, dragging point *D* (line 7) provided the impression of changing the relationship since point P then moved on a vertical line, and for the instrumentalisation process, dragging point *D* was not in line with the intentions of the task. It was necessary for the evolution of the instrumentation and instrumentalisation processes that the students understood the constraint of dragging point *D* indicating another relationship than the intended one. Ivy and Aya did not continue with the attempt

to investigate the conversions between point *P* and the area of the rectangle by dragging point *D*. Thus, for dragging to be an instrument, the students develop schemes that relate treatments within the individual representations and conversions between the point *P* and the area of rectangle *DABC* as results of their dragging. For example, they expressed this relationship between point *P* and the rectangle by making conversions to the symbolic or linguistic registers (line 10).

The term "intersection" is a word adopted from its appearance together with point *P* in the algebra view. This is a constraint of using GeoGebra as a recourse for natural language along with the symbolic register. Ivy's and Aya's use of intersection is a result of their instrumentation process, which led to inappropriate language in the situation and, potentially, to an incorrect understanding of point *P*.

Refining the design principles based on the 'coordinated' analysis

Based on our analyses, we refine the design principles and discuss how the networked theoretical perspectives have influenced the development of refined design principles using networking practices.

For HDC 1, the students use natural language throughout the task. At the beginning of the students' instrumentation, they express their interpretations of the representations and the properties of the rectangle. Through their communication, their interpretations develop. As they are expected to give a written answer, they express the discovered relationship using the natural language that appears in the algebra view. In this case, this is unfortunate. However, if the tool's expression is in line with the task, it can support the students' use of natural language.

For HDC 2, the students only investigate the rectangle at the beginning of the task sequence. In the transcript, Ivy and Aya also investigate the rectangle on their individual computers, when dragging and discussing that both the width and height of the rectangle influence the area (lines 2-6). Hence, the treatments in the rectangle are related to point *P* (lines 1, 7, 10). About the word "intersection", it is also important to be aware of the vast information provided in the algebra view. Hiding the algebra view could cause less confusion for the students to only focus on the graphical view.

For HDC 3, Ivy and Aya explore the template by dragging both points *A* and *D* at the beginning of their instrumentation process. When they drag, they can test their conjectures, e.g., regarding what happens to the rectangle when dragging either point *A* or point *D*. Through dragging, the students explore and investigate the relationships between the representations.

Applying the analyses to van den Akker's (1999) structure, the design principle may be defined as: If you want to design teaching with a digital tool for exercising and developing the representation competency in schools, then you are best advised to *gradually relate all four registers (linguistic, symbolic, figurative and graphic) in a task sequence when using digital tools.* Collaboration between students is essential for all three characteristics and procedures. This may be realised by:

1. including the linguistic register (both written and oral) throughout the task (i.e., when formulating the task, exploring the environment and representations, and writing answers), and being aware of how natural language appearing in the environment relates to the intentions of the task, as the linguistic register is often neglected in a digital tool, although it

is an important register expressing the relations between representations, formulating interpretations of different representations and for developing an artefact into an instrument.
2. breaking objects and tool windows into smaller units by introducing different representational registers and components of the artefact gradually (e.g., hiding the algebra view), because gradually introducing representations makes it possible to identify the specific characteristics of a representation and to help the students' processes of instrumental genesis by limiting their explorations to specific parts of the tool.
3. using sliders, dragging and tracing for explorations about different representations and the treatments and conversions caused by the dynamic features since these features make it possible to quickly move and translate between representations, and for developing conjectures about the representations' reciprocal relations.

Discussion of the potentials for networking of theories in DR

Reflecting on our work as networking of theories, the empirical case guided our work. Concerning the eight networking strategies (Prediger et al., 2008), and the compatibility of the theoretical perspectives, the two theoretical perspectives' backgrounds will be discussed. The background for the semiotics register approach may be defined as "Understanding mathematics is a cognitive process, which involves 'coordination of at least two registers'" (Duval, 2017, p. 110). The background for the instrumental approach we define as 'Using digital tools in mathematics is a cognitive process of developing suitable schemes for using the artefact'. Both theoretical perspectives are cognitive in a similar way as KOM. Hence, the perspectives are not too far apart and the networking process is identified as 'coordinating'. The theoretical perspectives have different objects, i.e., semiotic representations (Duval, 2017) and using digital tools for instrumental genesis (Trouche, 2005). Yet, the perspectives are possible to network as the process of instrumental genesis takes place through the representations present in the involved digital tool, and the students' understanding of conversion and treatments of the involved representations. The instrumental approach is informed by the semiotic register approach concerning how students understand and relate mathematical representations, related to mathematical understanding and activity (Duval, 2017). The semiotic register approach is enlightened by the use of digital tools and students' instrumental genesis (Trouche, 2005). The understanding of the students' use of the tool, while working with several mathematical representations, thus becomes deeper by coordinating the theoretical perspectives. The design principles reflect both theoretical perspectives. Yet, the design is only illustrated for one case.

In this paper, we have aimed to investigate how practices from networking of theories could strengthen the development of theory in DR. The implementation of networking practices in DR helps when having a situation involving two perspectives, which for our study is the use of digital tools and mathematical representations. The strategies of 'coordinating' and 'combining' fit the aims of DR with regard to understanding an empirical case and developing theories (Prediger et al., 2008; Prediger, 2019). Both in DR and networking of theories, the processes of analysis are determined by the objects of research. It is of course possible to conduct DR without the implementation of networking practices. Still, for a case such as ours, the relationships between the involved theories are not necessarily taken deeply into account related to each other. However, if aiming at using networking practices, such as 'coordinating' or 'combining', parallel analyses and investigation of

the compatibility of the theoretical perspectives involved are important. Implementing networking practices into DR, thus appear to strengthen the development of theories and theoretical perspectives.

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